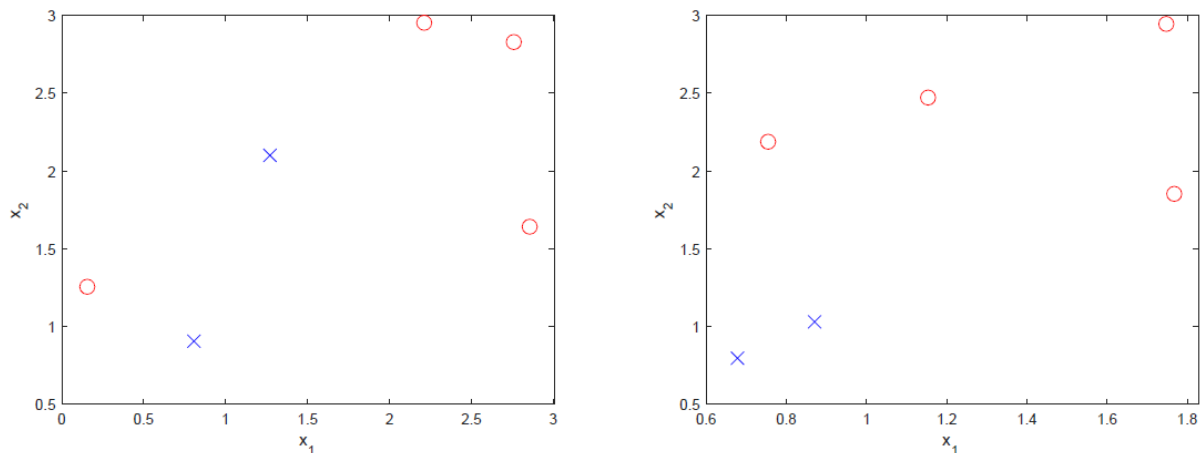


Linear Classification - Exercises

Exam: 04/07/2019

1. [Linear classification]

Consider two different datasets (see the left and right figures below) and the PLA algorithm to train a perceptron:



- 1.1 Describe the perceptron model.
- 1.2 Under which assumptions the PLA is guaranteed to converge? Provide a formal proof of convergence in such conditions.
- 1.3 Does the learning procedure terminate in the two cases illustrated in the figures?
- 1.4 If so, can we say how many steps we require to reach convergence?

1.1. See lecture notes.

1.2. The perceptron learning algorithm is guaranteed to converge in finite time if data are linearly separable. In such case, we are able to reduce the classification error to zero, otherwise the optimization procedure does not stop.
See lecture notes for the mathematical proof of convergence.

1.3. In the first case (left) we are sure that the algorithm does not converge. In the second case (right) the online procedure will eventually converge in finite time.

1.4. We cannot say anything about convergence rate, since it depends on the starting point for the parameters and the ordering of the points considered for training the model.

Linear Regression - Exercises

Exam: 04/02/2019

1. [Linear regression]

Consider a simplified learning scenario. Assume that the input dimension is 1, the data consists of 2 points $\{x_1, x_2\}$ and the target function is $f(x) = x^2$. We want to compute the constant model $h(x) = H$ that best fits the given points.

- 1.1 Formulate and solve the learning problem via linear regression (i.e., write the cost function and the solution of the least squares formulas).
- 1.2 What is the average model $\bar{h}$ (with respect to all possible experiments) when the points $\{x_1, x_2\}$ are drawn independently from a normal distribution with mean 0 and variance 1?

1.1. Ideal objective: min. wrt $h(x)$ out-of-sample error $E_{out}(h)$.

Resort to in-sample error

$$E_{in}(h) = \frac{1}{N} \sum_{n=1}^N \underbrace{(y_n)}_{f(x_n)} - \underbrace{h(x_n)}_H)^2 = \frac{1}{N} \sum_{n=1}^N (x_n^2 - H)^2 = \frac{1}{2} (x_1^2 - H)^2 + \frac{1}{2} (x_2^2 - H)^2$$

$$\operatorname{argmin}_H E_{in}(h) \Rightarrow \frac{\partial E_{in}(h)}{\partial H} = 0 \Rightarrow -(x_1^2 - H) - (x_2^2 - H) = 0$$

$$2H - x_1^2 - x_2^2 = 0 \Rightarrow H = \frac{x_1^2 + x_2^2}{2}$$

1.2. H_p. $x \sim \mathcal{N}(0, 1)$

$$\bar{h} = \mathbb{E}_x [h(x)] = \mathbb{E}_x \left[\frac{x_1^2 + x_2^2}{2} \right] = \frac{1}{2} \underbrace{\mathbb{E}[x_1^2]}_{\substack{\text{var}(x_1) \\ 1}} + \frac{1}{2} \underbrace{\mathbb{E}[x_2^2]}_{\substack{\text{var}(x_2) \\ 1}} = \frac{1}{2} + \frac{1}{2} = 1$$

Logistic Regression - Exercises

Exam: 04/02/2019

1. [Logistic regression]

You are working on a spam classification system using logistic regression. "Spam" is a positive class ($y = +1$) and "Not spam" is the negative class ($y = -1$).

1.1 Write the expression of the logistic regression model. How such a model can be used for classification?

1.2 You have trained your classifier with $N = 1000$ samples. The in-sample confusion matrix is:

	Actual Class: +1	Actual Class: -1
Predicted Class: +1	85	890
Predicted Class: -1	15	10

Compute the classifier precision, recall and F1-score.

1.3 Would you be satisfied by such results? Why? What would you do to improve the performance?



The output of such model represents the probability that a mail is classified as "spam". Logistic regression model can be used for classification by simply setting a threshold ϵ on the output.

1.2. Precision = $\frac{TP}{TP + FP} = \frac{85}{85 + 890} \approx 0.087$ ("how many predicted spam e-mails correspond to real spam?")

Recall = $\frac{TP}{TP + FN} = \frac{85}{85 + 15} = 0.85$ ("how many real spam e-mails are selected as spam?")

F1-Score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{0.07395}{0.937} \approx 0.158$

1.3. Precision and F1-Score much lower than 1 (there's a lot of false positive predictions) $\rightarrow$ Not satisfied.

* SPAM = $85 + 15 = 100$ * NOT SPAM = $890 + 10 = 900$

Reason of not satisfying performance: Unbalanced dataset

2. [Logistic regression]

Suppose we collect data for a group of workers with the following variables: hours spent working x_1 and number of completed projects x_2 . The workers may receive a bonus or not (represented by the binary variable y) depending on their overall performance. We fit a logistic regression model using the sigmoidal function

$$\sigma(w_0 + w_1 x_1 + w_2 x_2) = \frac{e^{w_0 + w_1 x_1 + w_2 x_2}}{1 + e^{w_0 + w_1 x_1 + w_2 x_2}}$$

and produce the following estimated coefficients: $w_0 = -6$, $w_1 = 0.05$, $w_2 = 1$.

- 2.1 Estimate the probability that a worker who worked for 40h and completed 3.5 projects gets a bonus.
- 2.2 How many hours would that worker need to spend working to have a 50% chance of getting a bonus?
- 2.3 Do you think that values of z in the logistic function $\sigma(z)$ lower than -6 make sense in this problem? Why?

$$2.1. \quad P(y=1|x) = \sigma(w_0 + w_1 x_1 + w_2 x_2)$$

$$x_1 = 40, \quad x_2 = 3.5$$

$$P(y=1|x) = \sigma(-6 + 0.05 \cdot 40 + 1 \cdot 3.5) = \sigma(-0.5) = \frac{e^{-0.5}}{1 + e^{-0.5}} \approx 0.3775$$

$$2.2. \quad P(y=1|x) = 0.5$$

$$\sigma(w_0 + w_1 x_1 + w_2 x_2) = \sigma(0)$$

$$w_0 + w_1 \hat{x}_1 + w_2 x_2 = 0 \Rightarrow -6 + 0.05 \cdot \hat{x}_1 + 1 \cdot 3.5 = 0 \Rightarrow \hat{x}_1 = 50h.$$

Alternative

$$P(y=1|x) = \frac{e^{w^T x}}{1 + e^{w^T x}} = 0.5 \Rightarrow e^{w^T x} = 0.5 + 0.5 \cdot e^{w^T x}$$

$$0.5 e^{w^T x} = 0.5 \Rightarrow e^{w^T x} = 1 \Rightarrow w^T x = \ln(1) = 0$$

$$w_0 + w_1 \hat{x}_1 + w_2 x_2 = 0$$

$$2.3 \quad z = \underbrace{w_0}_{-6} + \underbrace{w_1 x_1}_{>0} + \underbrace{w_2 x_2}_{>0} \geq 0 \geq -6$$

k-NN - Exercises

Exam: 11/01/2019

1. [k-NN]

A k-NN model is being used in a real-estate company to predict house prices.

- 1.1 Describe briefly the k-NN classifier and list the main differences with the perceptron.
- 1.2 Discuss how k should be selected depending on the size of the dataset.
- 1.3 A training set was used to generate a specific k-NN model with $k = 5$. The unseen observation below is presented to the model.

# of bedrooms	# of bathrooms	Size [m2]	Garage	House Price [€]
2	2	167	No	?

The k-NN model determines the observations in the table below from the training set to be the most similar. What would be the predicted house price value?

# of bedrooms	# of bathrooms	Size [m2]	Garage	House Price [€]
2	2	168	No	355000
2	2	142	Yes	352000
2	3	154	No	349000
3	2	202	No	356000
3	2	142	No	351000

1.1 Parametric vs Non-Parametric Classification.
See the lecture notes.

$$1.2 \quad k(N) = \lfloor \sqrt{N} \rfloor$$

1.3. House Price = 352 600 €.

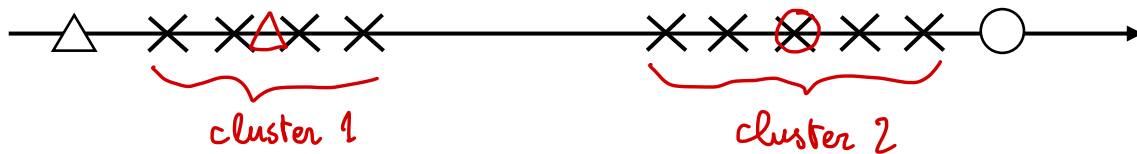
Since $K=5$, according to the theory, the prediction will be the average of the most similar (i.e. closest points). In this case, given the table, the predicted house price will be the mean of the house price

K-Means - Exercises

Exam: 11/01/2019

1. [Clustering]

- 1.1 Define the clustering problem. Is it a supervised or an unsupervised learning problem? Why?
- 1.2 Introduce the cost function for the K-means clustering problem and describe the Lloyd iterative algorithm.
- 1.3 Mark approximately in the graph below the solution (clusters and centers) found by Lloyd algorithm for the 2-cluster ($K = 2$) problem, when the data ($x_i \in \mathbb{R}$) are the crosses in the figure below and the algorithm is initialized with center values indicated with the triangle (cluster 1) and the circle (cluster 2).



1.1 See lecture notes.

$$1.2 \quad \min E_{IN} \left(\underbrace{S_1, \dots, S_K}_{\text{Partition in } K \text{ clusters}}, \underbrace{\mu_1, \dots, \mu_K}_{\text{vectors containing the coordinates of centroids}} \right) = \sum_{n=1}^N \|x_n - \mu(x_n)\|^2$$

- i) Initialization of centroids (greedy approach)
- ii) Identification of the closest observations
- iii) Update of centroids until E_{IN} stops decreasing.

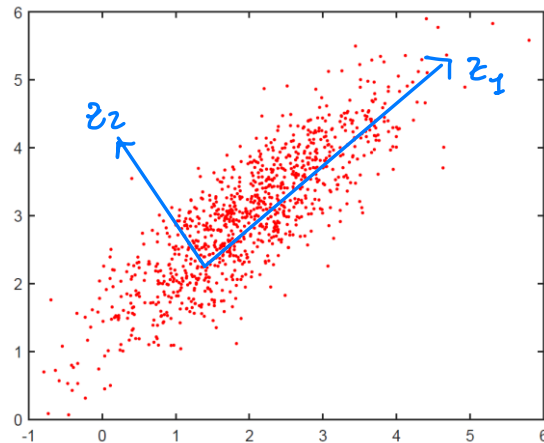
1. The Lloyd algorithm will find the two clusters. The centroids are moved to the centers in only one iteration, then the algorithm stops.

Data Pre-Processing - Exercises

Exam: 13/06/2019

1. [Data pre-processing]

Considering the following dataset.



1.1. Draw the direction of the principal components and say whether the dimensionality of the data could be reduced.

1.2. Apart from PCA, how can data be preprocessed in order to be best prepared for learning?

1.1. The variance of the data along dimension z_1 is larger than that along direction z_2 , thus a reduced 1-D feature space could be considered, depending on the specific problem.

1.2 See lecture notes.

- ① Input Centering
- ② Input Normalization
- ③ Input Whitening
- ④ Outliers removal

Neural Networks - Exercises

1. [Neural Networks]

Draw the scheme of a multi-layer feedforward neural network and explain how it overcomes the modeling limits of a simple perceptron.

See lecture notes.

Stochastic Processes - Exercises

1. [Stochastic processes]

Considering a zero mean stationary stochastic process $y(t)$.

- 1.1. Define its covariance function and show that it is an even function.
- 1.2. Discuss how to estimate the mean and the covariance function of $y(t)$ based on a set of N measurements $\{y_1, y_2, \dots, y_N\}$.

See lecture note